

A given table **Courses**(*studentID*, *courseName*) stores records about students enrolled in different courses. Consider the fact that there are only two courses 'PDSA' and 'DBMS'. If the following two queries produce exactly the same result what can we conclude about the data?

- `SELECT studentID FROM Courses WHERE courseName = 'DBMS'`  
`UNION`  
`SELECT studentID FROM Courses WHERE courseName = 'PDSA'`
- `SELECT studentID FROM Courses WHERE courseName = 'DBMS'`  
`UNION ALL`  
`SELECT studentID FROM Courses WHERE courseName = 'PDSA'`

- ☐ All students who took PDSA never took DBMS.
- ☐ All students who took PDSA also took DBMS.
- ☐ Certain students who took DBMS never took PDSA.
- ☐ None of the above

For the given tables **Employee** and **Department** answer questions 2 to 8

| Employee |          |        |        |
|----------|----------|--------|--------|
| eid      | ename    | salary | deptid |
| E001     | Arun     | 50000  | D001   |
| E002     | William  | 45000  | D002   |
| E003     | Ramesh   | 45000  | D004   |
| E004     | Rekha    | 35000  | D003   |
| E005     | Syed     | 30000  | D002   |
| E006     | Gurpreet | 80000  | D003   |
| E007     | Susan    | 85000  | D004   |

| Department |            |          |
|------------|------------|----------|
| deptid     | dname      | building |
| D001       | Accounts   | B1       |
| D002       | Marketing  | B2       |
| D003       | Sales      | B3       |
| D004       | Production | B1       |

Figure 9: Table Employee and Table Department

How many rows will be fetched by the following query?

1 point

```
SELECT eid FROM Employee WHERE deptid = 'D003' AND salary > 35000
EXCEPT
SELECT e.eid FROM Employee e, Department d
WHERE e.deptid = d.deptid AND d.building = 'B3';
```

- ☐ 2
- ☐ 1
- ☐ 0
- ☐ None of the above

What will be the output of the following query?

1 point

```
SELECT e.eid FROM Employee e, Department d
WHERE d.deptid = e.deptid AND d.building IN('B2')
UNION ALL
SELECT eid FROM Employee WHERE deptid = 'D002'
EXCEPT ALL
SELECT eid FROM Employee WHERE deptid = 'D002';
```

- ☐ E002
- ☐ E005
- ☐ The output will be blank
- ☐ E005

How many rows will be printed as the result of the following query?

1 point

```
SELECT MIN(salary),deptid FROM Employee GROUP BY deptid
INTERSECT
SELECT e1.salary,e1.deptid FROM Employee e1, Employee e2
WHERE e1.salary > e2.salary;
```

- ☐ 0
- ☐ 2
- ☐ 3
- ☐ 4

Write the number of tuples fetched by the following query.

```
SELECT eid FROM Employee, Department
EXCEPT ALL
SELECT eid FROM Employee
EXCEPT ALL
SELECT eid FROM Employee;
```

1 point

What will be the output of the following query?

1 point

```
SELECT MIN(salary),deptid FROM Employee GROUP BY deptid
INTERSECT
SELECT e1.salary,e1.deptid FROM Employee e1, Employee e2
WHERE e1.salary > e2.salary;
```

- ☐ 30000          D002
- ☐ 35000          D003
- 45000          D004
- 35000          D003
- 50000          D001
- ☐ 30000          D002
- 50000          D001
- 45000          D004
- ☐ 35000          D003

Choose the appropriate option for the query given below.

1 point

```
SELECT salary FROM Employee WHERE deptid = 'D001' AND salary > NULL;
```

The output is:

- ☐ 50000
- ☐ The query has error, it will not execute
- ☐ The query does not fetch any record
- ☐ None of the above

What will be the output of the given query?

```
SELECT SUM(salary)
FROM Employee
WHERE salary > 35000
GROUP BY deptid
HAVING SUM(salary) > 50000 AND SUM(salary) < 100000;
```

1 point

Consider the two queries given below

1 point

1. 

```
SELECT SUM(salary),deptid
FROM Employee
WHERE salary > 45000
GROUP BY deptid;
```
2. 

```
SELECT SUM(salary),deptid
FROM Employee
GROUP BY deptid
HAVING salary > 45000;
```

Choose all the correct statements regarding these two queries.

- ☐ Query 2 fetches SUM(salary) and deptid of all departments where the salary of each employee is greater than 45000
- ☐ While calculating the sum, Query 1 considers only those employee's salary for whom the salary is greater than 45000.
- ☐ Query 1 has syntax error
- ☐ Query 2 has syntax error

Consider the following statements.

1 point

1. Predicates in the HAVING clause are applied after the formation of groups.
2. Predicates in the WHERE clause are applied before formation of groups.

Choose the correct option

- ☐ Statement 1 is true, Statement 2 is false
- ☐ Statement 2 is true, Statement 1 is false
- ☐ Statement 1 is false, Statement 2 is false
- ☐ Statement 1 is true, Statement 2 is true